Nuclear Magnetic Resonance (NMR) Qubits

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Nuclear magnetic resonance is a phenomenon resulting from the placement of molecules inside of a strong, constant magnetic field being disturbed by a weak, oscillating magnetic field.[2] When an ensemble of particles is inside of a strong, constant magnetic field and disturbed by a weak, oscillating magnetic field, the average of all the nuclei's spin begins to spin like a top.[1] This occurs at the Larmor frequency, which is the precession frequency. This motion peaks at the resonance frequency, which increases as the strength of the strong magnetic field increases. The weak field is introduced as RF pulses to manipulate the nuclei.[1]

The physical model used to analyze NMR qubits is the Spin-½ object. NMR quantum computing takes place using ensembles of particles, originally in liquid form (LSNMR) but more commonly now in solid form (SSNMR).[3] Each nucleus is treated as an individual spin-½ qubit, but the readout is typically an ensemble measurement of many identical nuclei rather than measurements of single nuclei individually. The spin of each nucleus, influenced by surrounding electrons, stores the qubit information. By measuring the individual spins of each nucleus, each acts as one qubit.[4]

Since the spins of each nuclues are the qubits in this computer, the model is that of an ensemble of spin-½ objects. Basic NMR theory yields that for *n* coupled spins, the Hamiltonian is calculated as a result of each molecule's response to external and internal forces: external forces include static external magnetic fields and externally applied RF pulses to manipulate qubits whereas internal is constituted of the interactions between different nuclear spins (a.k.a. spin-spin interactions or J-coupling), the Zeeman effect from strong, external magnetic fields, and dipole/quadrupole coupling[7][12][13]. In NMR quantum computing it is typically assumed that there is weak coupling[8], i.e.

$$|J_{ij}| \ll |\omega_i - \omega_j|$$

Thus, the terms sum together as:

$$H_{total} = H_{external} + H_{internal} = (H_{Static} + H_{RF}) + (H_{Zeeman} + H_{J-Coupling} + H_{DC} + H_{QC})$$
$$= [-\sum_{i=1}^{n} \gamma B_0 I_z^{(i)} + \sum_{i=1}^{n} \hbar\gamma_i B_1(t)(I_x^{(i)} cos(\omega_{RF}t + \phi) + I_y^{(i)} sin(\omega_{RF}t + \phi))]$$
$$+ [\sum_{i=1}^{n} \hbar\omega_i I_z^{(i)} + 2\pi\hbar \sum_{i<j} J_{ij} I_z^{(i)} I_z^{(j)} + d(3I_z^{(i)} S_z^{(i)} - I^{(i)} * S^{(i)}) + \eta Q(3I_z^{(i)2} - I^{(i)} * I^{(i)})]$$

Where:

ω_i = Larmor frequency of spin *i*

$I_z^{(i)}$ = Spin operator along the z-axis for qubit *i*

J_{ij} = Scalar coupling constant between spins *i* and *j*

$B_1(t)$ = Time-dependent RF field amplitude

ω_{RF} = Driving frequency

γ_i = Gyromagnetic ratio for qubit i

ϕ = Phase of pulse[9]

However, the above is not practical when working with systems utilizing NMR. Depending on the application at hand, different assumptions are made to suit the controllable/effective parameters. In the case of NMR quantum computing, they are as follows[10][11][12]:

1. Dipole coupling can be assumed to have minimal effect (or time average to 0)
2. Quadrupole coupling vanishes for spin-½ objects (representation via Pauli matrices results in 0)
3. No perturbations (i.e. no external magnetic field or RF pulses)
4. Secular approximations

Combining all of these, the Hamiltonian simplifies to the below shown form:

$$H = H_{Zeeman} + H_{J-Coupling} = \sum_{i=1}^{n} \hbar\omega_i I_z^{(i)} + 2\pi\hbar \sum_{i<j} J_{ij} I_z^{(i)} I_z^{(j)}$$

Where:

ω_i = Larmor frequency of spin *i*

$I_z^{(i)}$ = Spin operator along the z-axis for qubit *i*

J_{ij} = Scalar coupling constant between spins *i* and *j*[9]

This same structure is used in NMR quantum computing, resulting in the same Hamiltonian being utilized. The associated energy levels for the Hamiltonian are the sums of the eigenvalues of each of the terms that constitute H_{total}. These are given as:

$$E_{Zeeman} = \sum_{i=1}^{n} \hbar\omega_i(\pm \tfrac{1}{2}),\ E_{J-Coupling} = \sum_{i<j}(\pm \tfrac{1}{4})J_{ij} I_z^{(i)} I_z^{(j)}$$

$$E_{total} = E_{Zeeman} + E_{J-Coupling} = \sum_{i=1}^{n}(\pm \tfrac{1}{2})\hbar\omega_i + \sum_{i<j}(\pm \tfrac{1}{4})J_{ij} I_z^{(i)} I_z^{(j)}$$

If perturbations are introduced into the system, the external forces that are present dominate the system, resulting in the Hamiltonian reasonably approximating to that of exclusively the external terms[3]:

$$H \approx -\omega_1^I I_x - \omega_1^S S_x$$

Energy levels now are equivalent to the Hamiltonian:

$$E = -\omega_1(\pm\tfrac{1}{2}) - \omega_1(\pm\tfrac{1}{2})$$

In LSNMR quantum computing, qubits are physically implemented using the nuclear spin-½ states of atoms (typically ^1H or ^{13}C)[5] within a selected molecule. An ensemble of these molecules are suspended in a liquid solvent and placed in a strong static magnetic field. The logical states $|0\rangle$ and $|1\rangle$ are represented by the nuclear spin being aligned or anti-aligned with the magnetic field, respectively. Where aligning with the magnetic field is the lower energy state.[5] When selecting a molecule, researchers consider three major parameters. First, each qubit-carrying nucleus must have a distinct resonant frequency so it can be individually distinguishable and addressable. Secondly, the used nuclei need to have a strong J-coupling network for multi-qubit operations and entanglement. Lastly, the nuclei need long coherence times to maintain quantum information.

In an NMR quantum computer, initialization relies on creating pseudo-pure states rather than initializing qubits to a pure $|00\ldots0\rangle$ state. Operation at room temperature results in the natural state of the system being a highly mixed thermal equilibrium where nuclear spins are almost randomly distributed, with only a tiny fraction of a percent favoring the lower energy state.[6] The thermal density matrix is manipulated so that it takes the mathematical form[10]:

$$\rho \approx \frac{I}{2^n} + \varepsilon\Delta\rho$$

where I / 2^n represents the large, unobservable background and $\epsilon\Delta\rho$ is the small deviation that produces the measurable signal. By applying tailored RF pulse sequences and gradient fields, this deviation term is transformed into a pseudo-pure state.[6] Because the large background identity matrix is invariant under unitary gate operations and produces no observable NMR signal, the measurable dynamics of the system are dictated entirely by the deviation matrix, $\Delta\rho$. By engineering this deviation term to be proportional to the density matrix of a desired pure state (such as $|00\ldots0\rangle\langle 00...0|$), the system reproduces the expectation values of that pure state up to an overall scaling factor.

RF pulses tuned to the resonant frequency of the targeted qubit function as single-qubit logic gates by rotating the spin vector on the Bloch Sphere. In NMR quantum computing, different nuclei within a molecule can act as separate qubits, and interactions between neighboring spins allow multi-qubit operations to be implemented.[6] By carefully timing RF pulses, these spin interactions can be controlled to perform quantum logic operations.[1] However, the system must complete computations before decoherence occurs, since relaxation processes gradually destroy the quantum state.

Readout is achieved by a weak ensemble measurement which monitors the state of the spin system without changing it.[6] A final RF pulse tips the nuclear spins into the transverse plane. As these spins precess together, their changing magnetic field induces an oscillating electrical current. Because each qubit precesses at a slightly different frequency, applying a

Fourier transform to the signal separates it into frequency peaks. The amplitude and phase of these peaks correspond to the final quantum states of individual qubits.

To demonstrate a specific gate operation, consider the implementation of a single-qubit Pauli-X gate on a ^1H nucleus suspended in a liquid sample. The system is placed in a strong static magnetic field of approximately 11.7 T, with a Larmor resonance frequency of roughly 500 MHz. Because the natural thermal polarization at room temperature is extremely small, the system is first initialized into a pseudo-pure $|0\rangle$ state. To execute the X gate, an RF pulse oscillating at the 500 MHz resonant frequency is applied orthogonal to the static field. This pulse exerts a weak magnetic field B_1 of 10^{-3} T. The exact pulse duration τ required for a π-rotation (180°) is calculated using the gyromagnetic ratio for ^1H ($\gamma = 2.675*10^8$ rad s^{-1} T^{-1})[5] and the relationship[14]:

$$\theta = \gamma \times B_1 \times \tau$$

Plugging into the formula, we get a pulse duration τ of approximately 11.74 μs. Mathematically, this operation is represented by the Pauli-X matrix acting on the initial state vector which successfully flips the qubit from the $|0\rangle$ state to the $|1\rangle$ state.

The potential for this implementation of quantum computing to be useful in the future is fairly high. Given the long history of NMR and its applications for quantum computing[10], the technology has matured to a point such that portable NMR quantum computers currently exist, as sold by SpinQ. They sell multiple training computers: the Gemini Lab, Gemini Mini, and Gemini Mini Pro. These are marketed as ways for students and researchers to gain hands-on experience with quantum computers, programming/gates, and general quantum phenomena. The Gemini Lab as well as the Triangulum II are also viable for research purposes. The combination of portability, a design intended for education, and a relatively low cost for entry will likely result in these computers being used for education purposes but unlikely for anything else. It's important to note that the scalability of LSNMR is highly limited, with the highest experimentally demonstrated systems reaching only about 12 qubits[15]. This limitation arises because the signal-to-noise ratio decreases exponentially as the number of qubits increases, due to pseudo-pure state dilution. In contrast, SSNMR offers better scalability potential in principle, though practical implementation still faces significant challenges.

The Gemini Lab/Mini/Mini Pro are all 2-qubit computers whereas the Triangulum II is a 3-qubit computer. As evidenced by these small numbers, the computers struggle to scale. Alongside this, the latent properties of NMR quantum computing prevent entanglement from being exploited which prevents the main benefit of quantum computing from being harnessed. As a result, it is unlikely that NMR quantum computing will be used in most modern systems attempting to glean the benefits of quantum computing over classical computing but it is probable that it will be used as a stepping stone or introductory tool to future quantum engineers/programmers.

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